

Niraj Dhakal

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EDUCATION

University of Maryland Baltimore County

Baltimore, MD

Bachelor of Science, Computer Science (GPA: 3.667)

- Achievements: Cum Laude
- Coursework: Machine Learning, AI/Algorithms, Database Systems, Software Engineering, Malware Analysis

SKILLS

- **Languages:** Go, Python, TypeScript/JavaScript, SQL (PostgreSQL), C/C++
- **Systems & Distributed:** Multi-user/distributed architecture, WebRTC/WebSockets, async pipelines, REST APIs
- **AI/ML:** LLM integration (Ollama, hosted APIs), RAG architecture, prompt engineering, context window optimization, pgvector
- **Web Development:** Gin, Next.js, React, Prisma, JWT authentication
- **Infrastructure & Tools:** Docker, PostgreSQL, MongoDB, Git, cloud/hosted API migration, agile/scrum, SDLC

EXPERIENCE

Dogwood Gaming | Software Engineering Intern

Jan 2025 – Present

- Owned end-to-end design and implementation of an LLM-powered chatbot subsystem for a production AI marketing platform, as one of three student engineers with no senior engineering oversight, taking high autonomy from prototyping through deployment and boosting social media metrics by 6%.
- Designed a retrieval-augmented system using PostgreSQL hybrid full-text search and trigram similarity over game documentation and scraped competitor data, enabling faster and more accurate information lookup for the chatbot.
- Debugged and optimized performance-critical, CPU-only large language model inference with Qwen2.5:3B through Ollama, including a model quantization migration that reduced inference latency from 5+ minutes to 3.5 minutes; led an architectural migration to a hosted cloud LLM API that reduced end-to-end query latency to 5 seconds.
- Built persistent chat history in PostgreSQL with JSONB storage and group-scoped authorization for the chatbot, allowing users to resume conversations and view past interactions across sessions.
- Refactored authentication from LocalStorage to HTTP-only JWT cookies with multi-user authorization across 16+ REST API endpoints, mitigating CSRF and eliminating XSS-based token theft.
- Containerized the full-stack application with Docker, wrote deployment scripts, and set up weekly schema/data backups that enable quick rollbacks and reduce downtime.

Retriever Proposal Portal | Scrum Master/Developer

Sep 2025

- Developed a full-stack web application using Next.js, TypeScript, Prisma, and MongoDB, enabling UMBC students to discover and join faculty-led initiatives while streamlining project discovery for students and faculty.
- Built role-based access control for three distinct user types (students, stakeholders, admins), enabling secure project discovery and collaboration.
- Collaborated with faculty stakeholders to define requirements and iterate on features, resulting in the platform being presented to campus leadership as a potential university-wide solution.
- Managed iterative agile development and sprint cycles using Jira and Git-based collaborative workflows, including backlog prioritization, task decomposition, and progress tracking to support the software development life cycle and ensure timely delivery.

CMSC 201 | Teaching Assistant

Sep 2025

- Hosted weekly labs for 18 students, teaching Python, data structures, and coding fundamentals.
- Held weekly office hours and graded assignments, providing feedback on code quality and logic.

PROJECTS

LocalStream

github.com/Niraj-Dhakal/local-stream

- Built a peer-to-peer screen-sharing platform using WebRTC for media streaming and a Go WebSocket server for signaling, supporting 5+ concurrent, cross-device viewers over a local network with HTTPS.
- Implemented thread-safe room management to handle concurrent viewer connections and prevent race conditions in shared state, applying core distributed-systems and concurrency principles.
- Scaled the architecture from 1-to-1 to 1-to-many connections while maintaining direct P2P media paths and using the server solely for signaling, without media relay.